

B2 BACnet / Modbus

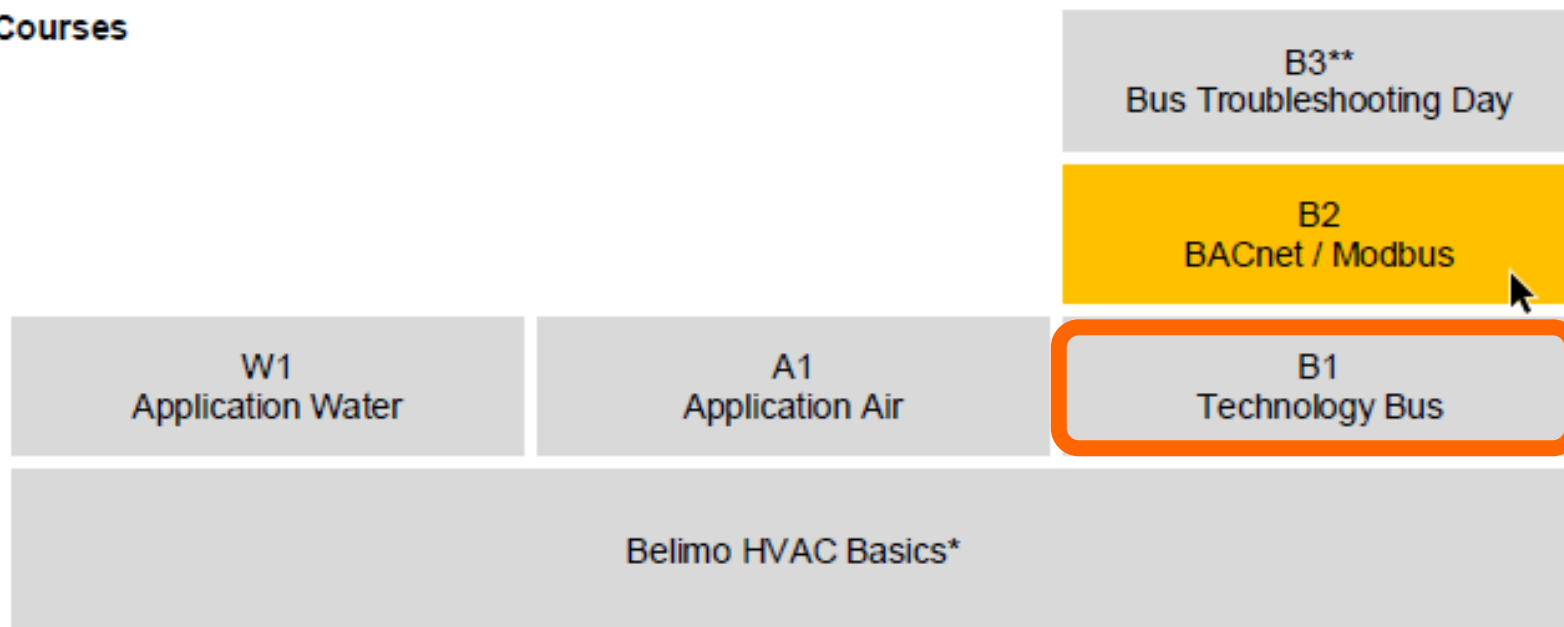
Modbus

March 4th – 5th, 2026

B2 BACnet / Modbus

Based on **B1 Online Learning Path**

Courses



* Part of the Belimo introduction program for new employees

** Recurring specialist meeting for 1st Level Support

Preparation



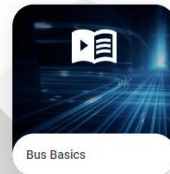
B1 Technology Bus Basics (en)

Lernfortschritt: 0%

Basic topics about bus communication
and bus protocols in buildings.

Start

In the first learning path step you
will find the course description of
this learning path.



Course Description

Course Description of the B1 Course

Los

Abschliessen

Feedback



Handouts

Bus@Belimo Bus Basics B1

Technology Box

Communicative Actuators

03.2025

For internal use only!

Product	Control / Bus Protocol										HybridMode	MP Profile	MP Client	Sensors	Tools	Cloud	Misc Functions													
	OIC	3-Point	0-10V	4-20mA	PWM	MP Bus 1)	Modbus RTU	Modbus TCP	BACnet MS/TP	BACnet IP	HTTP / REST	KNX	MP Bus	Modbus	BACnet	MP-Partner 2-3)	UK24MOD	UK24BAC	M Bus	Inputs	Passive	Active	Switch	PC-Tool	ZTH EU 4)	Belimo Assistant 2	Web Server		BACnet Cov Incl. Writeable	BACnet/Modbus Bus-Watchdog
General Air/Water	..MFT	•	•	•	•	8									Air/Water	•	•	•	•	1	•	•	•	•	•					
	..MFT2					8									Air/Water	•	•	•	•	1	•	•	•	•	•					
	..MP	•	•	•	•	8									Air/Water	•	•	•	•	1	•	•	•	•	•					
	..MPL					16									MPL	•	•	•	•		•	•	•	•	•					
	..MOD			•	•	8		•					•	•	Air/Water	•	•	•	•	1	•	•	•	•	•				•	•
	CQ...BAC					•	•	•																	•		Indirect	•	•	
	PR...BAC	•	•	•	•	8		•	•				•	•	Air/Water	•	•	•	•	2	•	•	•	•	•	•	Indirect	•	•	
..KNX												•	•	•					1	•	•	•	•	•	•					
VAV	..V-D3-MP		•	•	•	8									VAV/EPiV	•	•	•	•	1	•	•	•	•	•					
	..V-D3-LON																			1										
	..V-D3-MOD		•	•	•	8	•	•					•	•	VAV/EPiV	•	•	•	•	1	•	•	•	•	•				•	•
	..V-D3-KNX												•	•						1										
	VRU...BAC		•	•	•	8	•	•					•	•	Datapool	•	•	•	•							•		Indirect	•	•
En. Water	CMV...MP		•	•	•	16									VAV/EPiV	•	•	•	•							•		Indirect	•	•
	ZoneEase VAV					•																								
	EP...R+MP / P6...W...E-MP		•	•	•	8																				•				
	EP...R+MOD / P6...W...E-MOD		•	•	•	8																				•				
	EP...R2+BAC		•	•	•	8																				•				
Io T	EP...R-RS+BAC		•	•	•	16	•	•																		•				
	EV...R+BAC / P6...W...EV-BAC		•	•	•	16	•	•																		•		Indirect	•	•
	EV...R2+BAC / +MID		•	•	•	16	•	•								Datapool	•	•	•	•					•		direct	•	•	
																Datapool	•	•	•	•					•		direct			
	V...24A-LP1												•	•	•					2	•	•	•	•		•		direct		
Safety	BF24TL-T-ST					8									Fire Damper	•	•	•												
	BKN230-24-C-MP					8									Fire Damper	•	•	•												
	BKN230-24-LON						•	•																						
Sensor	BKN230-24-MOD																													
	22...15...						•	•																						
	22...16...							•																						
	22RT...1...1					16									Datapool	•	•	•	•							•				
Meter	22RT...1...2					16	•	•							Datapool	•	•	•	•							•				
	22PF...1U...		•			16	•	•	•	•	•	•	•	•	Datapool	•	•	•	•						•					
	22PE...1...		•			16	•	•	•	•	•	•	•	•	Datapool	•	•	•	•						•		direct			

For internal use only!

- 1) Max. number of nodes per trunk
- 2) See actual MP-Partner overview (www.belimo.com)
- 3) Datapool devices require separate implementation
- 4) ZTH offers full, limited support or is controlled by actuator (S)

Modbus RTU



MP  **BUS**
MP-BUS COMPATIBLE



 **Modbus**

TCP



RTU



 **BACnet**[™]

MSTP



IP



KNX[®]

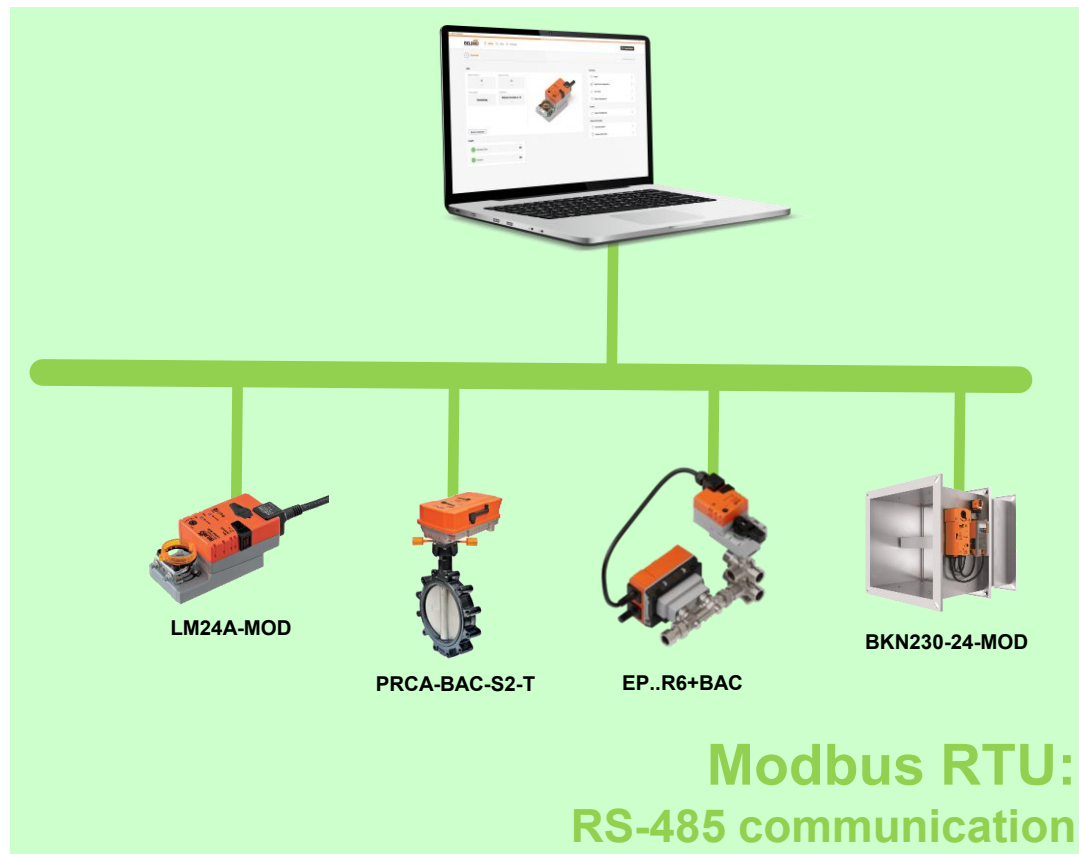
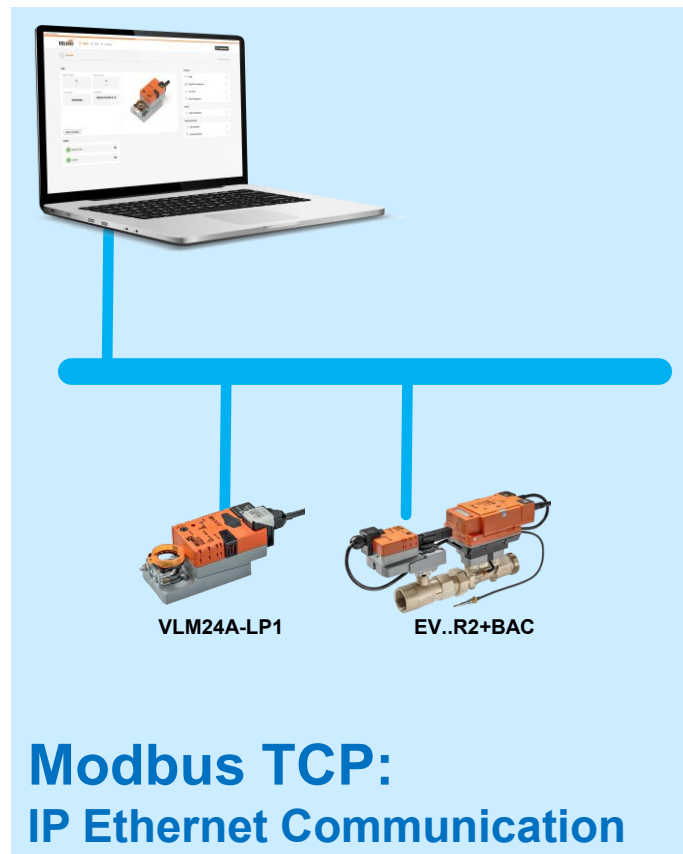


Terminology

Master / slave -> Client / server

- Gradually updating documents with the new terminology
- Politically no longer correct
- Historically:
 - Modbus: **master / slave bus**
 - BACnet MS/TP: **Master Slave Token Passing**
- **New terminology:**
 - Modbus: **client / server bus**
 - BACnet MS/TP: **Manager Subordinate Token Passing**

Modbus TCP and Modbus RTU

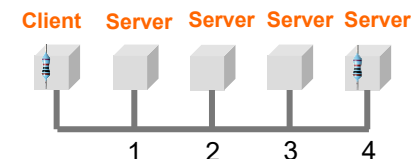


Modbus - General

Modbus	
Fields	Heating, Ventilation, Air Conditioning, Lighting, Fire alarm, Intrusion and Access Control Systems
Application Building	Modbus is widely used in building services engineering and industry. Modbus RTU is applied in the Automation- and Field Level .
System Setup	RTU: Client-Server (single-client / multi-server) TCP: Client-Server
Actuator / Sensor	RTU: Server TCP: Server
Medium	RTU: RS485/EIA485 (max. length per line without repeater 1'200m)
Topology	RTU: Bus line with terminating resistor
Max. Number of participants	Typically 32 devices per line Device-specific limits and possibilities check Modbus interface description!
Speed	9'600 - 115'000 Baud, often 38'400 Bd, network with low performance
Norms, Standards Links	De-facto standard http://www.modbus.org/ http://modbus.org/docs/Modbus over serial line V1_02.pdf
Strengths	- Simple and cheap implementation - Widespread
Weaknesses	- No standard - Values must be interpreted



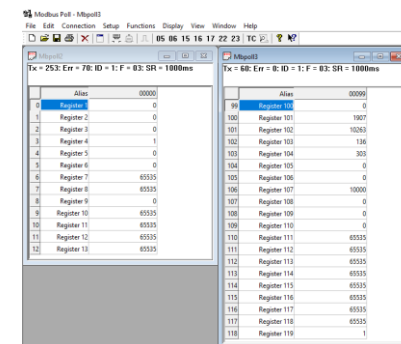
Bus Line:



Modbus Interface Description:



Tool Modbus Poll:



Modbus Register and Functions

Modbus Server

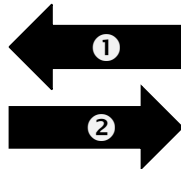
Actuator / Sensor

Modbus Register:

Rel. Setpoint % (Register Nr. 1)
Rel. Position % (Register Nr. 5)
Setpoint Source (Register Nr. 119)

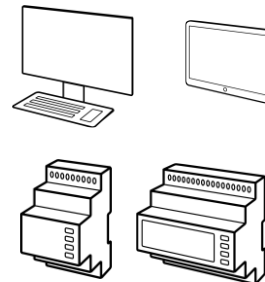


Modbus Functions:
Read Holding Register (03)
Write Single Register (06)
Write Multiple Register (16)



Modbus Client

Automation station / Management station



The Modbus server (e.g. actuator) has the data which can be queried by the Modbus client (e.g. automation station) via functions.

The data is "raw" and must be interpreted correctly.

Modbus Register and Interpretation – Example: NM24A-MOD

Modbus Poll - Mbpoll2

File Edit Connection Setup Functions Display View Window Help

Tx = 407; Err = 70; ID = 1; F = 03; SR = 1000ms

Alias	Value
Register 1	10000
Register 2	0
Register 3	0
Register 4	1
Register 5	343
Register 6	3
Register 7	65535
Register 8	65535
Register 9	0
Register 10	65535
Register 11	65535
Register 12	65535
Register 13	65535

Tx = 214; Err = 0; ID = 1; F = 03; SR = 1000ms

Alias	Value
Register 101	0
Register 102	0
Register 103	303
Register 104	0
Register 105	0
Register 106	0
Register 107	10000
Register 108	0
Register 109	0
Register 110	0
Register 111	65535
Register 112	65535
Register 113	65535
Register 114	65535
Register 115	65535
Register 116	65535
Register 117	65535
Register 118	65535
Register 119	1

Setpoint Relative in % (Register Nr. 1)

Register Nr. 1 = 10'000

Description of Register Nr. 1 in the Modbus-Register documentation:

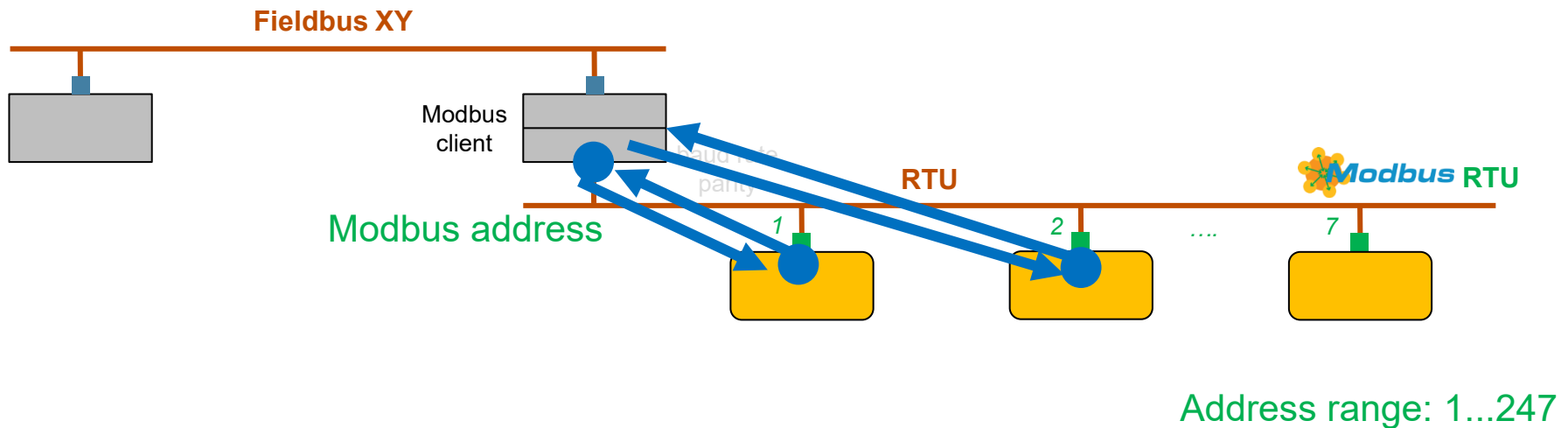
Modbus Register Description						
No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W

With the description you can interpret the value as follows:

$10'000 * \text{Scaling Factor} = 100.00 \%$. In addition, we see that register no. 1 is the setpoint value which is writeable.

Modbus RTU

- **1...247** Modbus address
- **0** used for Broadcasts



Modbus RTU Addressing

RS-485 Serial Communication

Modbus Client – e.g. Modbus Poll



Every device connected to the same line must have the same baudrate and the same parity!

Baud rate = 19'200 (bits per second - velocity of the bytes running through the cable)
Parity = 1-8-N-2 (how checks of the bits are done)

Count up server address from 1:

Addr = 1



Modbus gateway

Addr = 2



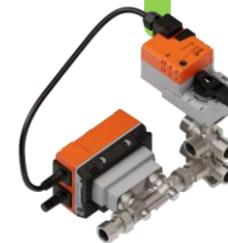
LM24A-MOD

Addr = 3



PRCA-BAC-S2-T

Addr = 4



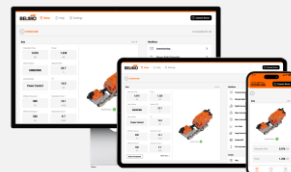
EP..R6+BAC

Addr = 5



BKN230-24-MOD

Parameterisation tools



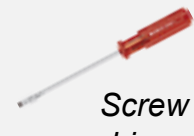
Belimo Assistant 2



LINK.10



ZTH EU



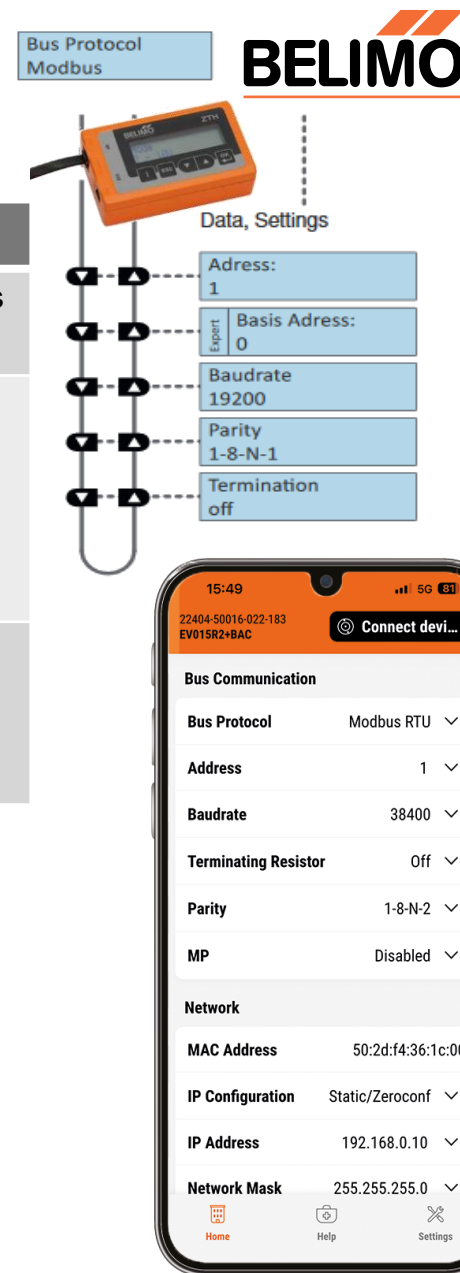
Screw driver

Modbus RTU - Configuration

Modbus RTU Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, unique in each Modbus Line
Baud rate	9'600 19'200 38'400 76'800 115'200	Baud rate of the RS485 Network . Insert the Terminating Resistor at the first and last device of the line. Supported baud rates of the specific device can be found in the Register Description.
Parity	1-8-N-2 1-8-N-1 1-8-E-1 1-8-O-1	Transmission format must be set the same on the whole bus line, including the client and all servers.

➡ Check max. number of devices per Modbus Line (typically 32 devices)



LM24A Modbus Register Description



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LM24A-MOD



Rotary actuator, 5 Nm, AC/DC 24 V, BACnet MS/TP, Modbus RTU, MP-Bus, 2...10 V, 150 s (35...150 s), IP54

List price CHF 338.00

1

Add to Cart

Add to Project List

Share

Downloads

Accessories

Details

Documentation

- ☐ **Technical data sheet – LM24A-MOD** [Download](#)
Technical data sheet | English | 789 KB | pdf
- ☐ **Installation instructions - NM...A... / LM...A... / VC** [Download](#)
Installation instructions | 652 KB | pdf
- ☐ **EU Declaration of Conformity - LM24A-MOD** [Download](#)
EU Declaration of Conformity | 28 KB | pdf
- ☐ **Brochure – Bus solutions from Belimo** [Download](#)
Product overview | English | 2044 KB | pdf
- Product and Price Catalogue** [View now](#)
Product and price catalogue
- ☐ **Tool Connections** [Download](#)
Guide / Manual | 1430 KB | pdf

Tender texts

CAD / BIM

System integration

Current documents

- ☐ **BACnet Interface description - Air / water actuators (for application software version 3.08)** [Download](#)
Interface description | English | 592 KB | pdf
- ☐ **Introduction to MP-Bus Technology** [Download](#)
Guide / Manual | English | 4 MB | pdf
- ☐ **Modbus Interface description - Air / water actuators (for firmware version 3.08)** [Download](#)
Interface description | English | 611 KB | pdf

Archive

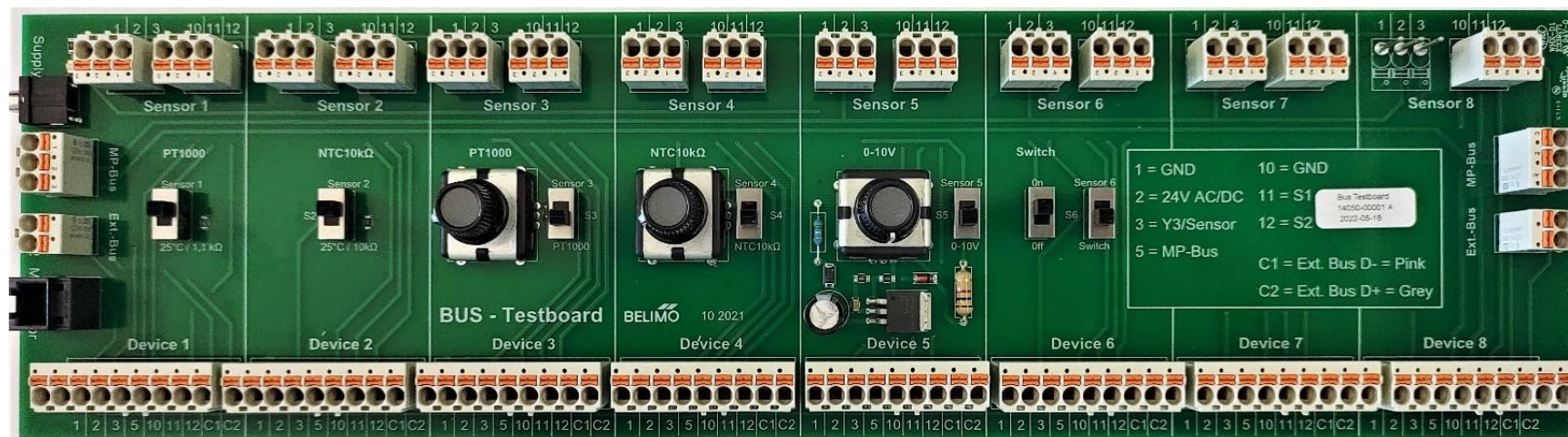
Binary numbers



Binary numbers

- Placeholder -> PPT

Workshop Introduction



Workshop Introduction



- Placeholder -> PPT

Workshop I

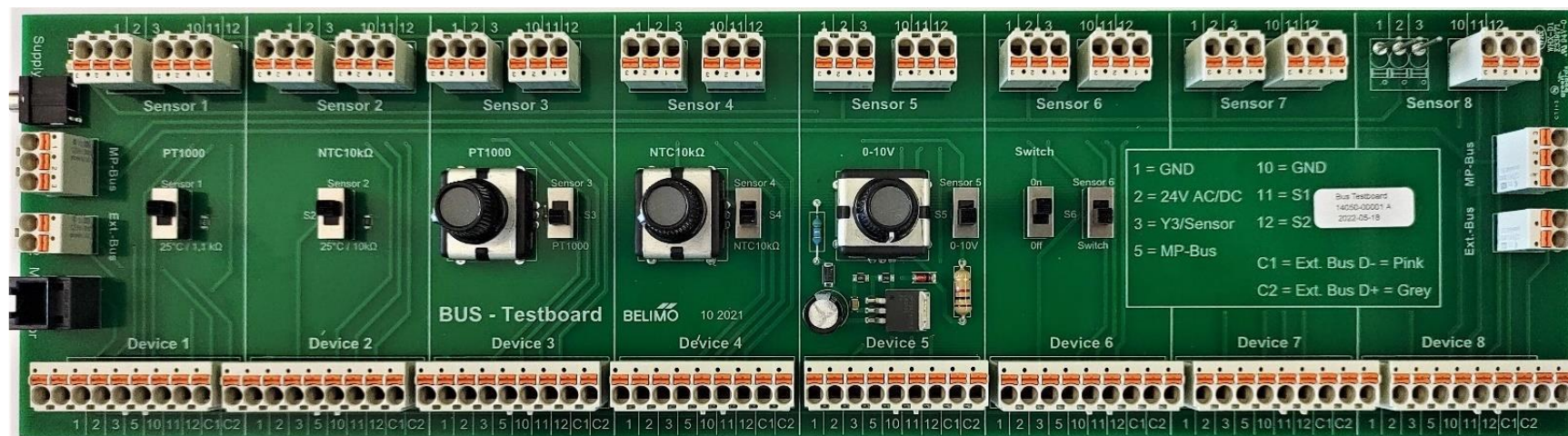


Workshop I



- Placeholder -> PPT

Workshop II



Workshop II



- Placeholder -> PPT

Group picture



IT network basics



IT network basics



- Placeholder -> PPT

Modbus TCP





TCP

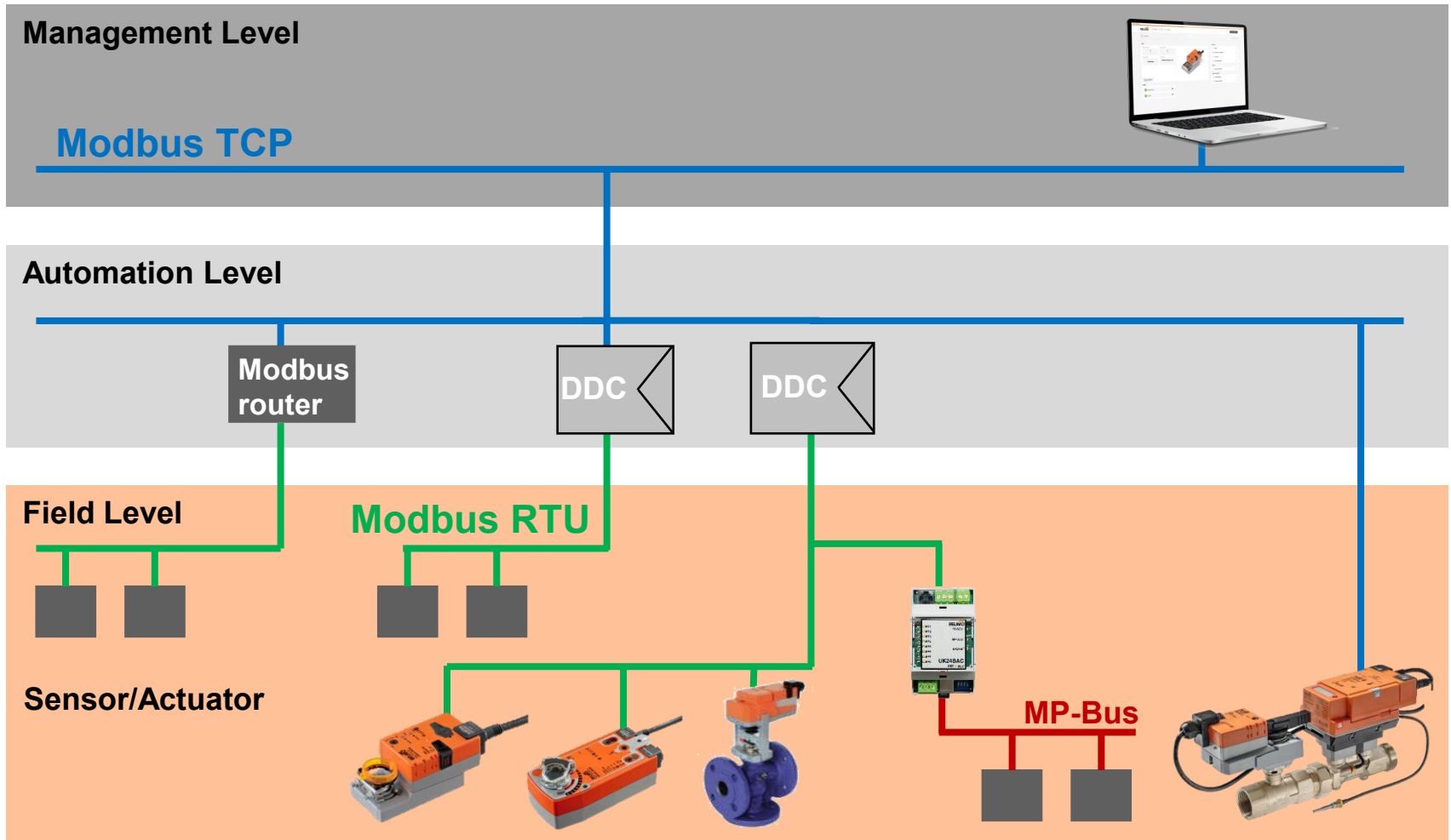
RTU

MSTP

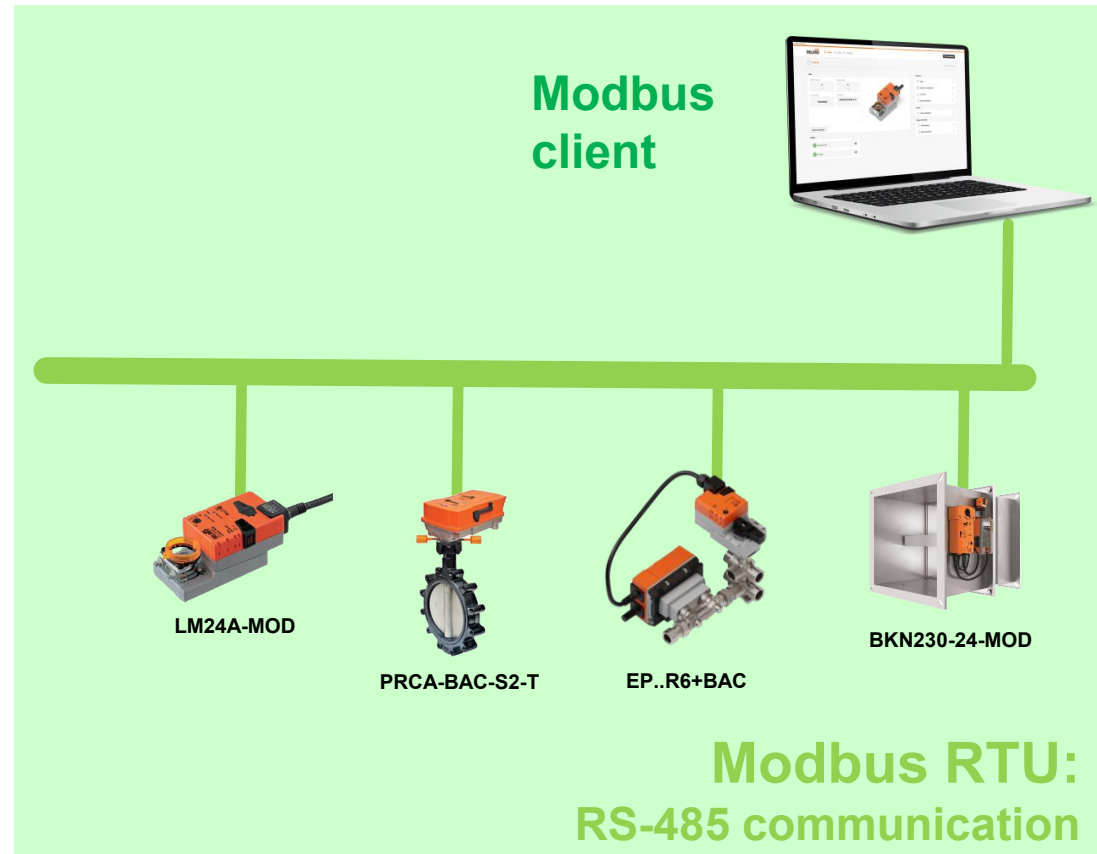
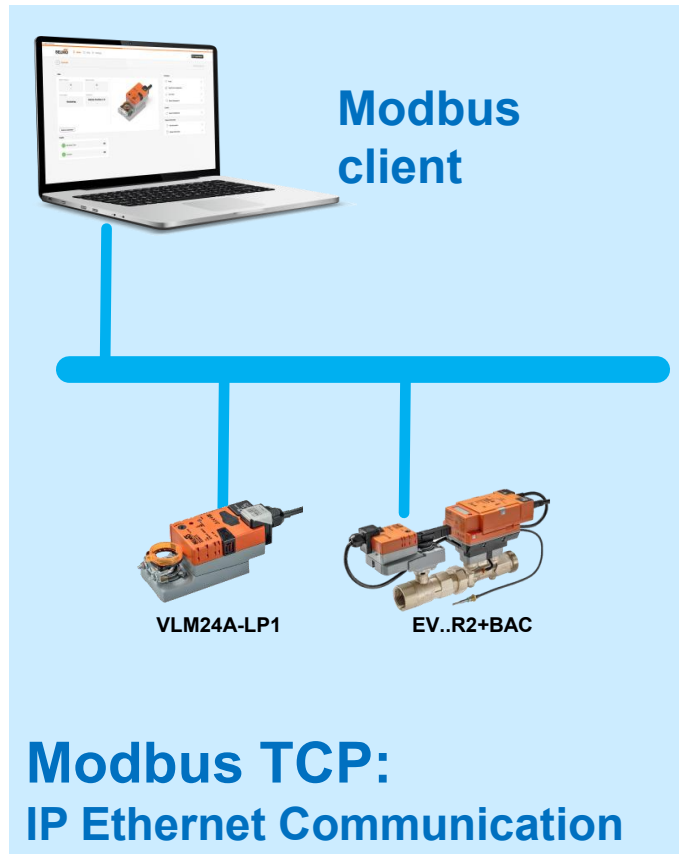
IP



Modbus TCP and Modbus RTU



Modbus IP and Modbus MS/TP



Modbus TCP Addressing



IP Subnet: 192.168.0.1...254
Subnet Mask: 255.255.255.0



Modbus client

Port: 502, Timeout = 30s

Addr: 1



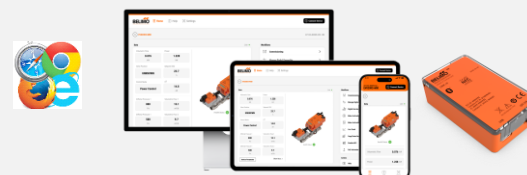
VLM24A-LP1

Addr: 2



EV015R2+BAC

Parameterisation

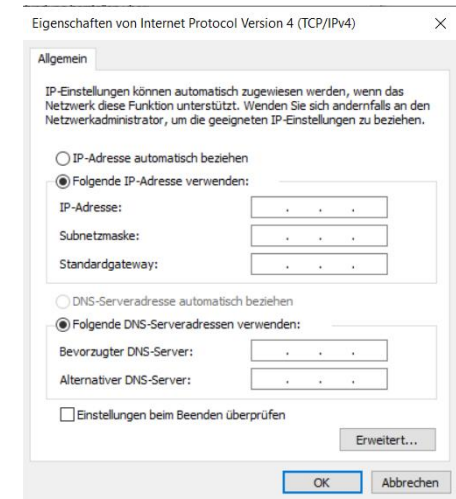


IP - Configuration

IP Configuration for Belimo Devices

IP Address	For example: 192.168.1.10	IP Address , corresponding to the IP Subnet (IP V4 address)
Subnet Mask	For example: 255.255.255.0	Subnet Mask of the IP Ethernet network (Definition of Subnet Size)
Default Gateway	For example: 192.168.1.50	Default Gateway of the IP Ethernet subnet (Address to which all packets with IP addresses from other subnets should be sent: IP Router to other networks)

➔ *IP Addressing must be configured according to the specific IT Network*



Modbus TCP - Configuration

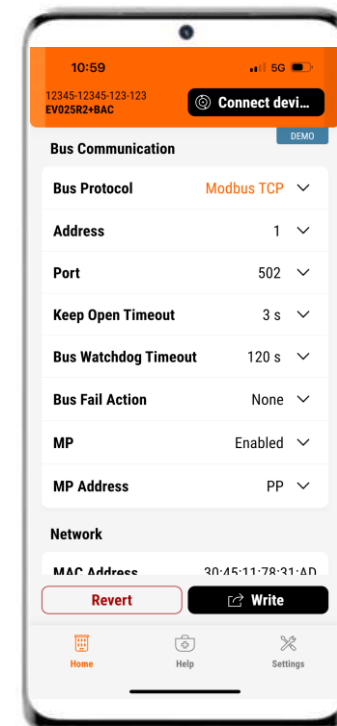
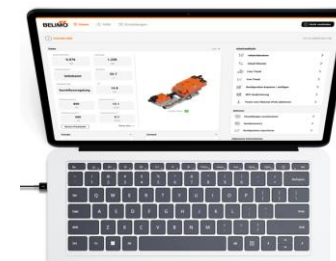
Modbus TCP Configuration for Belimo Devices

Modbus Address	1-247	Modbus Address, is only used for TCP in case that the Modbus device is connected to a Modbus RTU network. The IP Address for TCP Communication is configured in the IP settings of the device.
Port	Default: 502	Can be selected free. Default is 502.
Timeout	1-99	Seconds the actuator keeps the connection open. Typically 30-60s (default 30s)

Modbus TCP Settings

1	Modbus Address
502	TCP Port
60	Timeout [seconds]

BELIMO

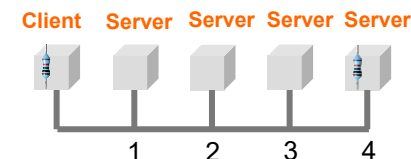


Modbus - General

Modbus	
Fields	Heating, Ventilation, Air Conditioning, Lighting, Fire alarm, Intrusion and Access Control Systems
Application Building	Modbus is widely used in building services engineering and industry. Modbus RTU is applied in the Automation- and Field Level .
System Setup	RTU: Client-Server (single client) TCP: Client-Server
Actuator / Sensor	RTU: Server TCP: Server
Medium	RTU: RS485/EIA485 (Max. length per Line without Repeater 1200m)
Topology	RTU: Bus Line with Terminating Resistor
Max. Number of participants	Typically 32 Devices per Line Device-specific limits and possibilities check Modbus Interface Description!
Speed	9'600 - 115'000 Baud, often 38'400 Bd, Network with low performance
Norms, Standards Links	De-facto Standard http://www.modbus.org/ http://modbus.org/docs/Modbus over serial line V1_02.pdf
Strengths	- Simple and cheap implementation - Widespread
Weaknesses	- No standard - Values must be interpreted



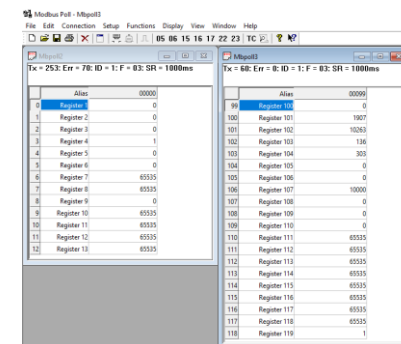
Bus Line:



Modbus Interface Description:



Tool Modbus Poll:



Modbus Register and Functions

Modbus Server

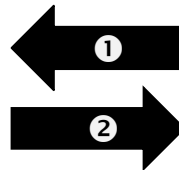
Actuator / Sensor

Modbus Register:

Rel. Setpoint % (Register Nr. 1)
Rel. Position % (Register Nr. 5)
Setpoint Source (Register Nr. 119)

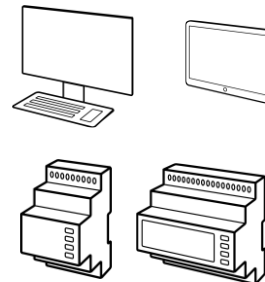


Modbus Functions:
Read Holding Register (03)
Write Single Register (06)
Write Multiple Register (16)



Modbus Client

Automation station / Management station

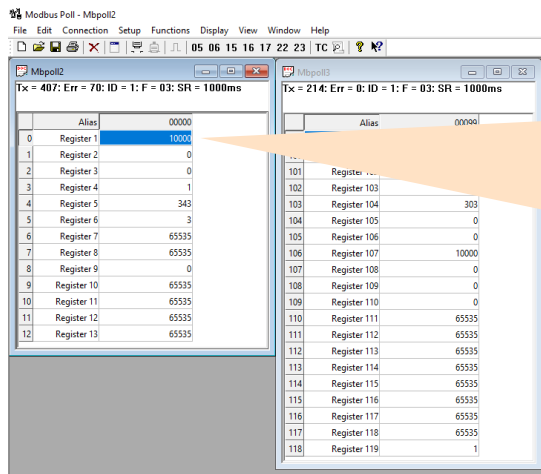


DDC

The Modbus server (e.g. actuator) has the data which can be queried by the Modbus client (e.g. automation station) via functions.

The data is "raw" and must be interpreted correctly.

Modbus Register and Interpretation – Example: NM24A-MOD



Setpoint Relative in % (Register Nr. 1)

Register Nr. 1 = 10'000

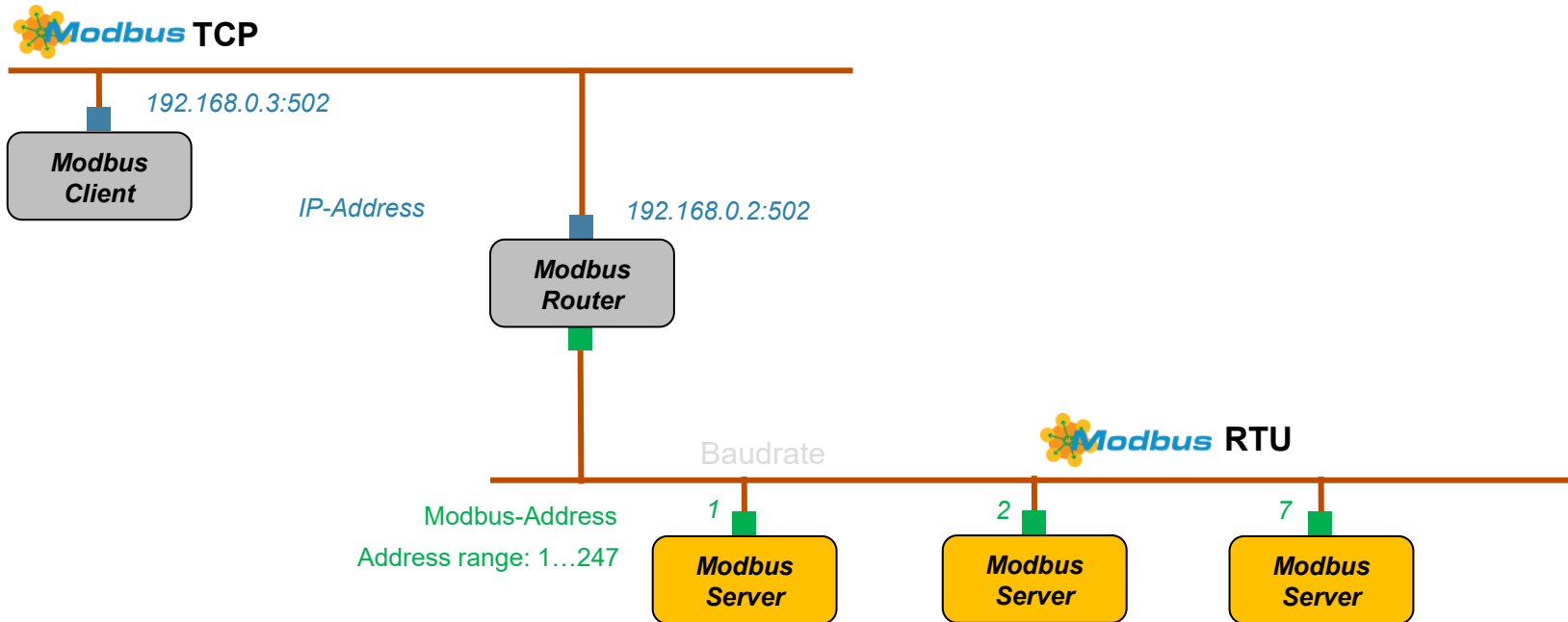
Description of Register Nr. 1 in the Modbus-Register documentation:

Modbus Register Description

No.	Address	Description Comment	Range Enumeration	Unit	Scaling	Access
1	0	Setpoint Setpoint for actuator between Min (Register No. 106) and Max (No. 107)	0...10000 Default: 0	%	0.01	R / W

With the description you can interpret the value as follows:

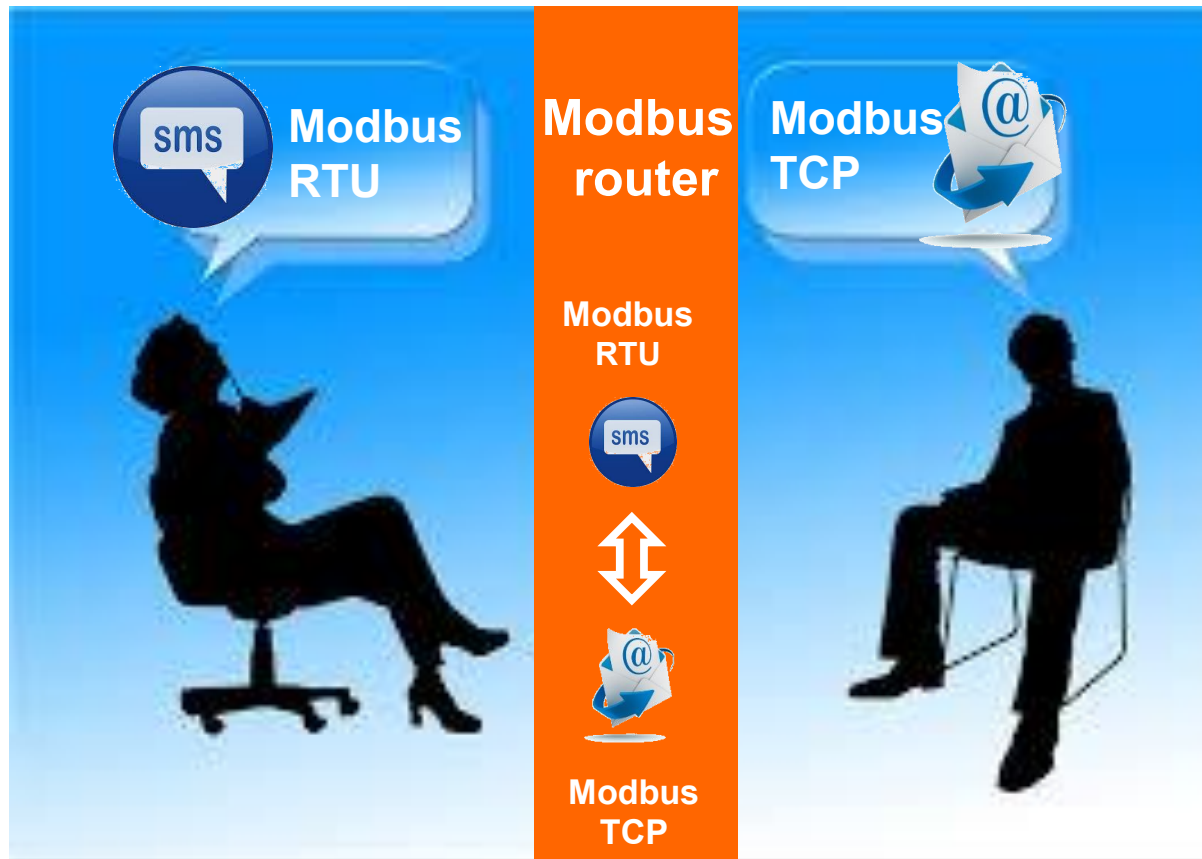
$10'000 * \text{Scaling Factor} = 100.00 \%$. In addition, we see that register no. 1 is the setpoint value which is writeable.



Transport / Transmission: **Router**

BELIMO

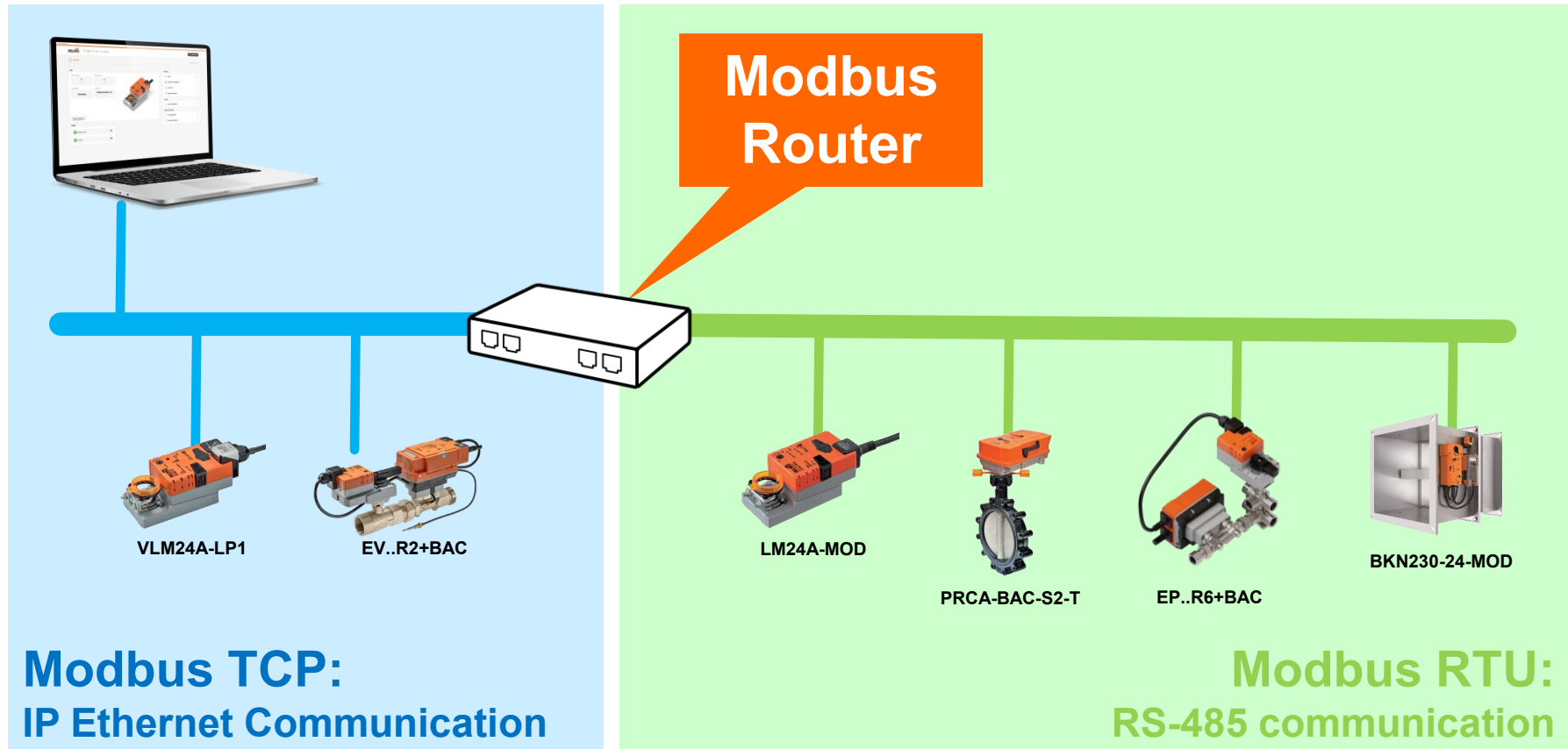
Connection of different Transport Options



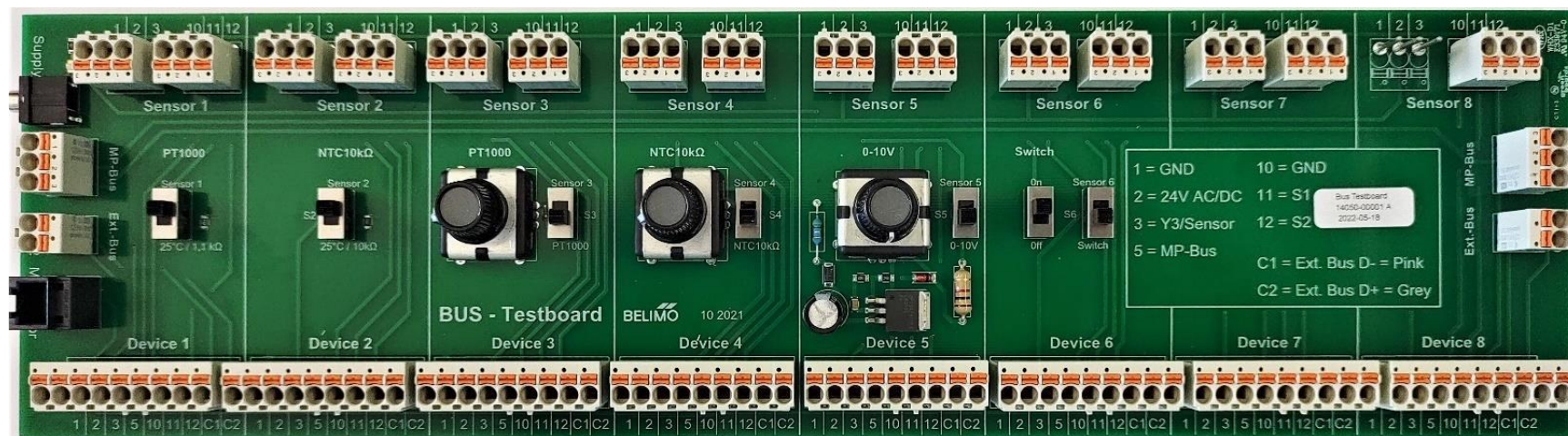
Modbus TCP and Modbus RTU



➔ *Modbus addresses must be unique in the entire Modbus line!*



Workshop III



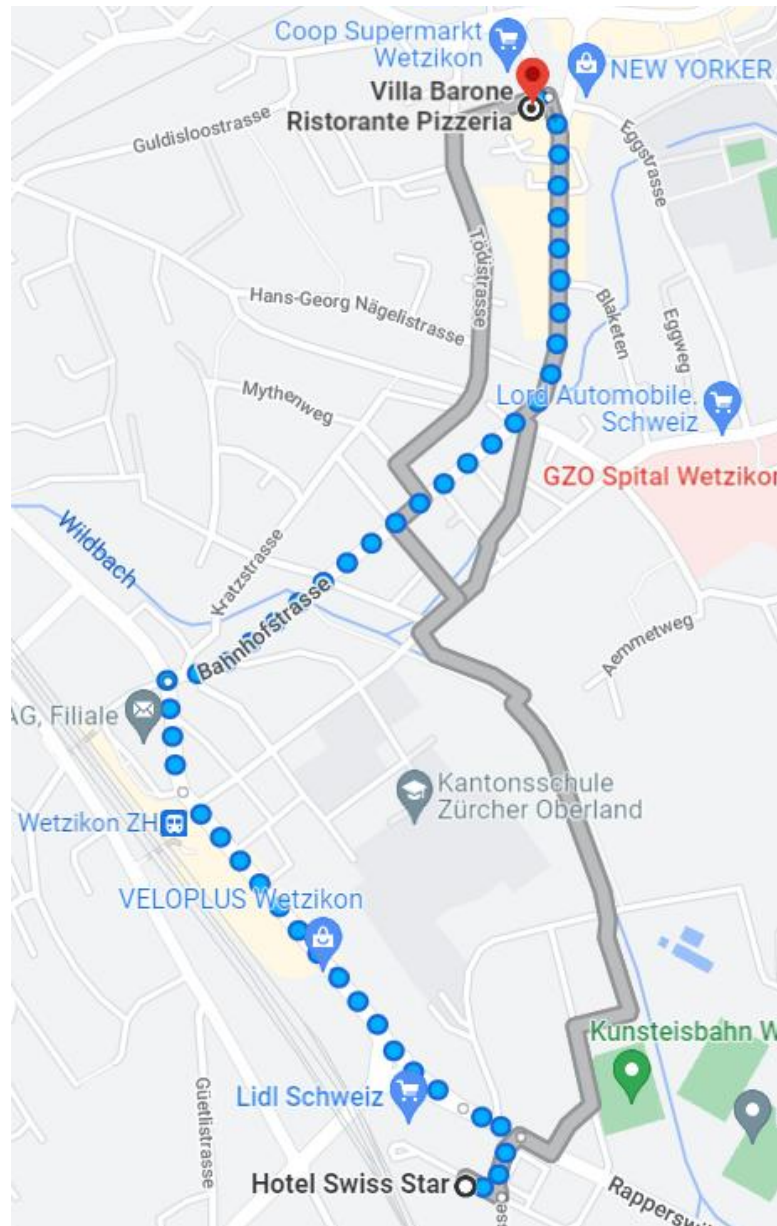
Workshop III



- Placeholder -> PPT

Dinner

- Today 18:30
- <https://www.villa-barone-wetzikon.ch/>
- Meeting 18:10 at hotel entrance



Tomorrow



- Start 08:30



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